

Comprehensive Acoustic Analysis Software for Transverse Flutes

Advanced Tools for Geometric Modeling, Impedance Analysis, and Acoustic Characterization

Your Name

Your Institution

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Outline

- 1 Introduction
- 2 Geometric Visualization
- 3 Acoustic Analysis
- 4 Resonator Truncation Analysis
- 5 Sensitivity Analysis
- 6 Database & Statistics
- 7 Engineering Tools
- 8 Multi-Flute Comparison
- 9 Technical Implementation
- 10 Applications & Use Cases
- 11 Future Directions
- 12 Conclusion

- **Comprehensive software suite** for acoustic analysis of transverse flutes
- **Integration** with OpenWind library for impedance computation
- **Multi-flute comparison** and statistical analysis capabilities
- **Geometric modeling** with support for cylindrical and conical holes
- **Advanced acoustic metrics** for instrument characterization

Key Capabilities

- Geometric visualization (2D profiles, 3D models, engineering drawings)
- Acoustic impedance/admittance analysis
- Resonator truncation analysis (novel method)
- Sensitivity analysis for geometric parameters
- Database management and statistical reporting

Core Components:

- FluteData / FluteDataDB
 - Data loading and validation
 - Geometry concatenation
 - Database caching
- FluteAnalyzer
 - Acoustic metrics calculation
 - Multi-flute comparison
- ResonatorTruncationAnalyzer
 - Novel truncation analysis

Visualization & Tools:

- FluteOperations
 - 2D/3D plotting
 - Engineering drawings
- FluteGeometryEditor
 - Interactive geometry editing
- SensitivityAnalysis
 - Parameter variation studies

Unified PyQt5 GUI integrates all components

Multiple visualization modes:

① Combined Profile

- Internal acoustic profile (cork-referenced)
- External physical profile (with wall thickness)
- Hole positions and dimensions

② Individual Parts

- Headjoint, left, right, foot sections
- Internal and external profiles per part

③ Axial Cross-Section

- 2D view with hole polygons
- Support for cylindrical and conical holes

④ Top View

- Concentric circles for conical holes
- Hole positions along the bore

Hole Geometry Support

Cylindrical Holes:

$$D_{\text{hole}} = \text{constant}$$

Conical Holes (Undercut):

$$D_{\text{outer}} < D_{\text{inner}}$$

The system automatically:

- Calculates wall thickness at hole position
- Generates conical geometry for OpenWind
- Visualizes undercut angle: $\theta = \arctan\left(\frac{D_{\text{inner}} - D_{\text{outer}}}{2h}\right)$

Example JSON Format

```
"Holes diameter": [12.0, [14.0, 18.0], 9.0]
```

Cylindrical — Conical (outer, inner) — Cylindrical

3D Visualization

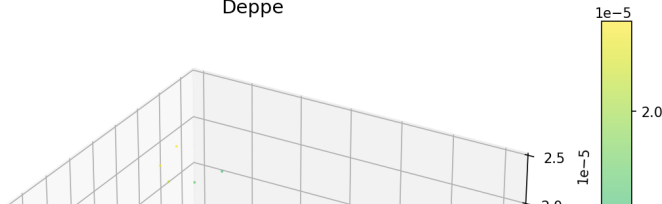
Interactive 3D Models:

- PyVista-based 3D rendering
- Internal and external geometry
- Multi-flute comparison in 3D space
- Rotatable, zoomable, interactive views

Features:

- Lazy loading (initializes only when accessed)
- Part-by-part visualization
- Color-coded sections
- Export capabilities

Mapa 3D: Frecuencia vs. Longitud vs. Amplitud
Deppe



Impedance and Admittance

Fundamental Quantities:

Acoustic impedance:

$$Z(\omega) = \frac{P(\omega)}{U(\omega)} = R(\omega) + jX(\omega)$$

Acoustic admittance:

$$Y(\omega) = \frac{1}{Z(\omega)} = G(\omega) + jB(\omega)$$

where:

- $P(\omega)$: Acoustic pressure
- $U(\omega)$: Volume velocity
- R, X : Resistance and reactance
- G, B : Conductance and susceptance

Computation: OpenWind's ImpedanceComputation using transmission line theory

Resonance Frequencies

Antiresonance Frequencies:

$$f_n = \arg \min_f |Z(f)| = \arg \max_f |Y(f)|$$

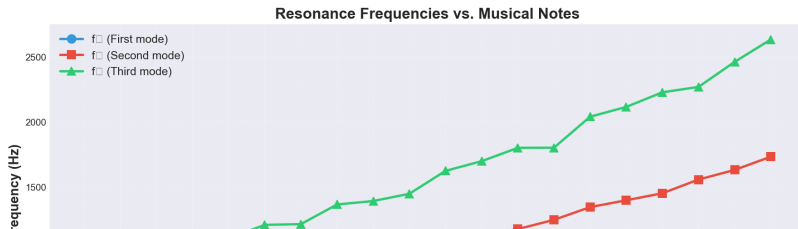
For an ideal open-open tube:

$$f_n = \frac{nc}{2L_{\text{eff},n}}$$

where:

- n : Mode number (1, 2, 3, ...)
- c : Speed of sound (≈ 343 m/s at 20°C)
- $L_{\text{eff},n} = L + \Delta L_{\text{head},n} + \Delta L_{\text{foot},n}$: Effective length

Visualization: Resonance frequencies vs. musical notes, compared to equal temperament



Inharmonicity

Definition:

$$\Delta = 1200 \log_2 \left(\frac{f_{\text{play}}}{f_0} \right) \quad [\text{cents}]$$

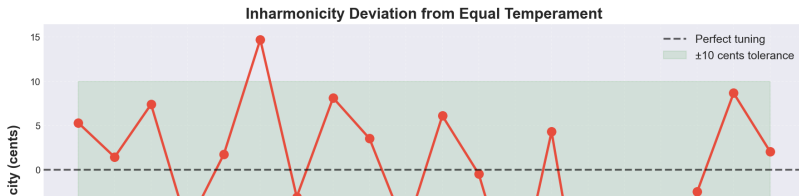
where:

- f_{play} : Target playing frequency (from fingering chart)
- f_0 : First antiresonance frequency (impedance minimum)

Interpretation:

- $\Delta \approx 0$: Well-tuned note
- $\Delta > 0$: Note is sharp (player must flatten)
- $\Delta < 0$: Note is flat (player must sharpen)

Physical Significance: Quantifies the intonation adjustment required relative to the instrument's passive resonance



Mouthpiece Octave Compensation (MOC)

Physical Basis: Frequency-dependent embouchure end correction:

$$L_{\text{eff},n} = L + \Delta L_{\text{head},n} + \Delta L_{\text{foot},n}$$

Headjoint cavity impedance:

$$Z_{\text{cav}}(f) = -jZ_0 \cot(kL_h)$$

where $k = 2\pi f/c$ and L_h is the headjoint cavity length.

Octave Ratio:

$$\frac{f_{11}}{2f_1} = \frac{L_{\text{eff},1}}{L_{\text{eff},2}} = \frac{L + \Delta L_{\text{head},1}}{L + \Delta L_{\text{head},2}}$$

Typically: $\Delta L_{\text{head},1} > \Delta L_{\text{head},2}$ (second mode sharp relative to octave)

MOC Metrics

Embouchure Length Corrections:

$$\Delta l_n = \frac{c}{4} \left(\frac{1}{f_{\text{res},n}} - \frac{1}{f_{\text{play},n}} \right)$$

MOC Difference:

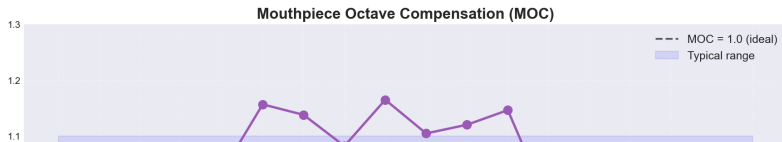
$$\text{MOC}_{\Delta} = \Delta l_I - \Delta l_{II} = \frac{c}{4} \left(\frac{1}{f_{\text{res},1}} - \frac{2}{f_{\text{play}}} \right)$$

MOC Ratio:

$$\text{MOC}_R = \frac{\Delta l_I}{\Delta l_{II}} = \frac{1 - \frac{f_{\text{res},1}}{f_{\text{play}}}}{1 - \frac{f_{\text{res},2}}{2f_{\text{play}}}}$$

Displayed Metric:

$$\text{MOC} = \frac{f_1 - f_0}{f_{\text{play},II} - f_{\text{play},I}}$$



First-Octave Pitch Adjustment (B_I)

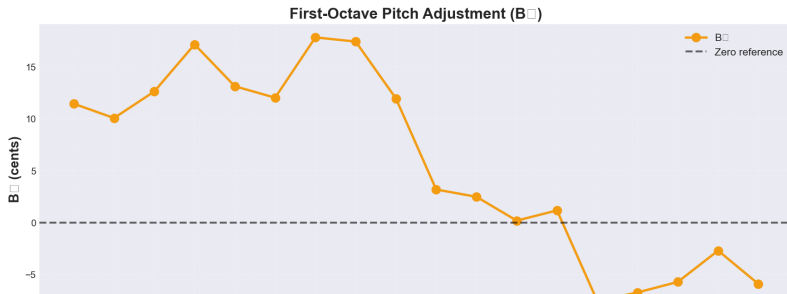
Definition:

$$B_I = 1200 \log_2 \left(\frac{f_{\text{play},I}}{f_0} \right) \quad [\text{cents}]$$

Physical Interpretation:

- $B_I > 0$: Player needs to play sharper than passive f_0
- $B_I < 0$: Player needs to play flatter than passive f_0
- $B_I \approx 0$: Passive f_0 is already close to target pitch

Significance: Directly reflects the basic intonation challenge presented by the instrument for each note in the first register.



Embouchure Shift Pitch Effect (ESPE)

Embouchure Corrections:

$$\Delta l_{\text{emb},n} \approx \frac{nc}{2} \left(\frac{1}{f_{\text{play},n}} - \frac{1}{f_n} \right)$$

$$\Delta l_{\text{I,emb}} = \frac{c}{2} \left(\frac{1}{f_{\text{play,I}}} - \frac{1}{f_0} \right)$$

$$\Delta l_{\text{II,emb}} = c \left(\frac{1}{f_{\text{play,II}}} - \frac{1}{f_1} \right)$$

Differential Correction:

$$\Delta(\Delta l_{\text{emb}}) = \Delta l_{\text{II,emb}} - \Delta l_{\text{I,emb}}$$

ESPE:

$$\text{ESPE} = 1200 \log_2 \left(\frac{L_{\text{eff,I}}}{L_{\text{eff,I}} + \Delta(\Delta l_{\text{emb}})} \right) \quad [\text{cents}]$$

Interpretation: Estimates the pitch change induced on the fundamental by the differential embouchure correction required for the octave jump.

Q-Factor

Definition:

$$Q = \frac{f_0}{\Delta f}$$

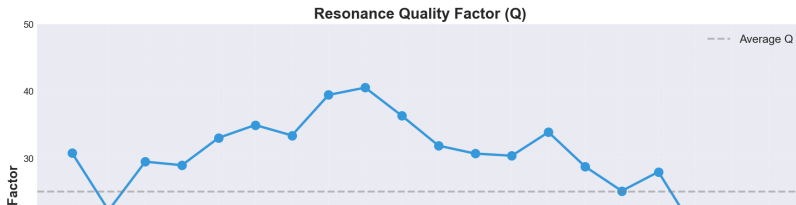
where Δf is the bandwidth at -3 dB from the peak admittance.

Calculation:

- Find peak admittance: $|Y|_{\max}$ at f_0
- Calculate threshold: $|Y|_{\text{threshold}} = |Y|_{\max}/\sqrt{2}$
- Find frequencies where $|Y| = |Y|_{\text{threshold}}$: f_{left} , f_{right}
- Bandwidth: $\Delta f = f_{\text{right}} - f_{\text{left}}$

Physical Significance:

- Higher Q : Sharper resonance, better mode isolation
- Lower Q : Broader resonance, more damping



Harmonic Ratios

Definition:

$$\text{Ratio}_{f_1/f_0} = \frac{f_1}{f_0}$$

$$\text{Ratio}_{f_2/f_0} = \frac{f_2}{f_0}$$

$$\text{Ratio}_{f_2/f_1} = \frac{f_2}{f_1}$$

Ideal Values (Perfect Harmonic Series):

- $f_1/f_0 = 2.0$ (octave)
- $f_2/f_0 = 3.0$ (twelfth)
- $f_2/f_1 = 1.5$ (perfect fifth)

Deviations indicate:

- Bore conicity effects
- End correction frequency dependence
- Hole interaction effects



Peak Heights and Tonal Characteristics

Admittance Peak Heights:

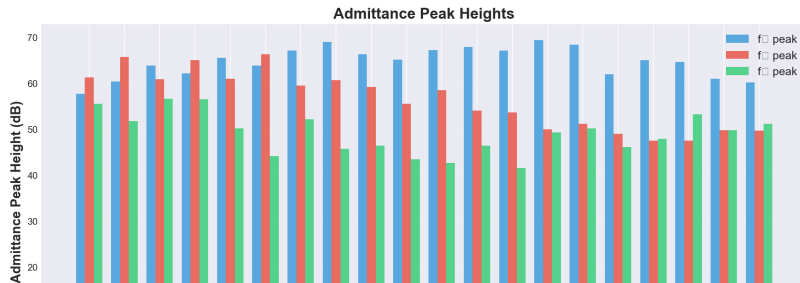
$$|Y|_{\text{dB}} = 20 \log_{10}(|Y|) \quad [\text{dB}]$$

Significance:

- Higher peaks: Stronger resonance, easier to excite
- Lower peaks: Weaker resonance, requires more player effort

Tonal Characteristics:

- Mode coupling strength
- Register transition ease
- Timbre quality indicators



Novel Analysis Method

Concept:

- Analyze the acoustic response of the **bore alone** (without holes)
- Progressively truncate the geometry from the end
- Study how length and conicity affect resonance frequencies

Advantages:

- **Isolates pure geometry effects:** No hole interference
- **Reveals length-frequency relationships:** Direct observation of how truncation affects modes
- **Acoustically valid:** Similar to "cutoff length" studies in wind instrument acoustics

Method:

- 1 Truncate geometry to percentage of total length (100%, 95%, 90%, . . . , 20%)
- 2 Calculate impedance for each truncated section
- 3 Extract acoustic metrics for each length
- 4 Visualize trends vs. truncated length

Truncation Geometry

Process:

- Maintain cork position at $x = 0$
- Truncate from the end (foot side)
- Interpolate diameter at truncation point if needed
- Ensure minimum length (default: 50 mm)

OpenWind Configuration:

- No side holes (except optionally embouchure)
- Dummy fingering chart (all holes "open")
- Source location: "entrance" (embouchure)
- Open end radiation: unflanged

Effective Length:

$$L_{\text{truncated}} = L_{\text{total}} \times \frac{\text{percentage}}{100}$$

Resonator Truncation Metrics

For each truncated length, we calculate:

- 1 **Resonance Frequencies:** $f_0, f_1, f_2, \dots$

$$f_n = \arg \min_f |Z(f)|$$

- 2 **Inharmonicity:**

$$\Delta_{\text{cents}} = 1200 \log_2 \left(\frac{f_1}{2f_0} \right)$$

- 3 **Harmonic Ratios:**

$$\frac{f_1}{f_0}, \quad \frac{f_2}{f_0}, \quad \frac{f_2}{f_1}$$

- 4 **Effective Length:**

$$L_{\text{eff}} = \frac{c}{2f_0}$$

- 5 **Q-Factor:** For each mode

Resonator Truncation Visualizations

Four Main Plots:

① Resonance Frequencies vs. Length

- Shows how f_0 , f_1 , f_2 change with truncation
- Reveals conicity effects on mode spacing

② Inharmonicity vs. Length

- Quantifies deviation from perfect harmonic series
- Shows how geometry affects octave tuning

③ Harmonic Ratios vs. Length

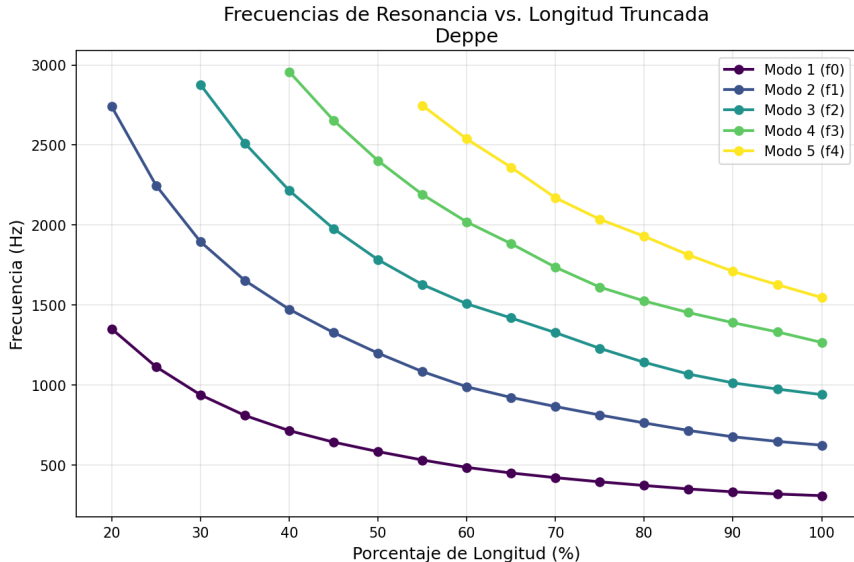
- f_1/f_0 and f_2/f_0 with ideal reference lines
- Reveals mode coupling and conicity effects

④ Impedance Curves Overlay

- Superimposed impedance magnitude for different lengths
- Shows frequency shift and peak broadening

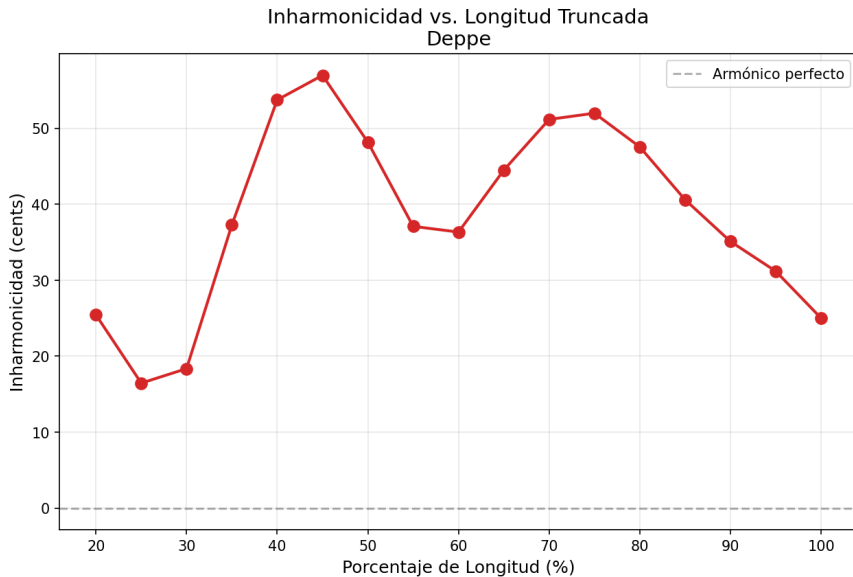
Note: See dedicated slides below for individual plots.

Resonator Truncation: Frequencies vs. Length

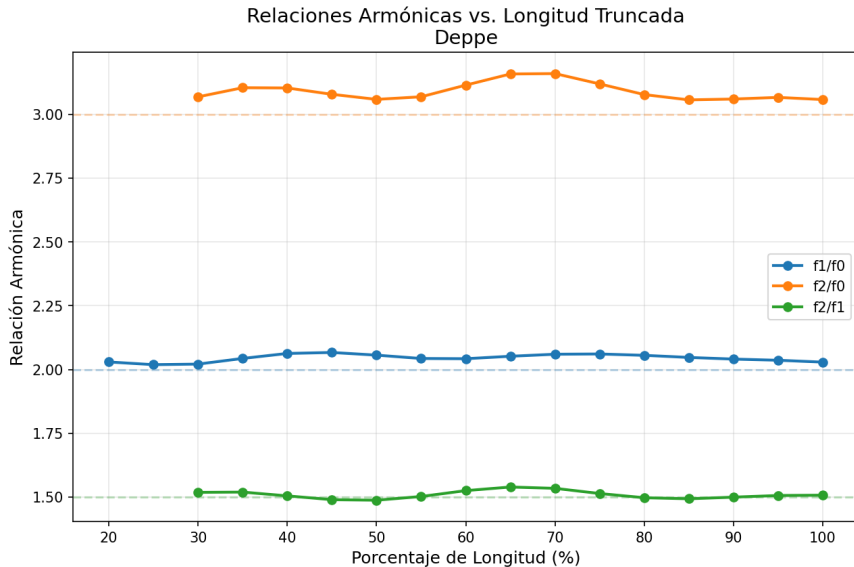


Shows how fundamental and overtone frequencies change with truncation

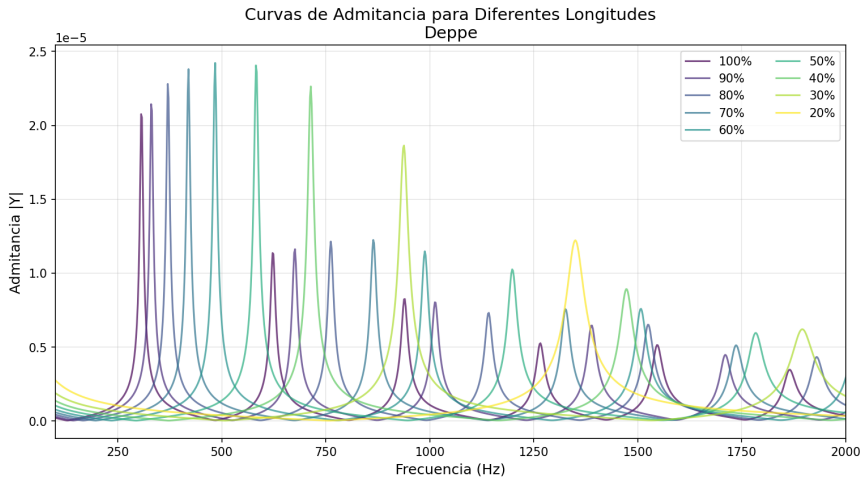
Resonator Truncation: Inharmonicity vs. Length



Resonator Truncation: Harmonic Ratios vs. Length



Resonator Truncation: Impedance Overlay



- Superimposed impedance curves for different truncated lengths
- Visualizes frequency shift and peak broadening with length reduction

Resonator Truncation: Physical Interpretation

Key Observations:

- **Length-Frequency Relationship:**

$$f_n \propto \frac{1}{L_{\text{truncated}}}$$

As length decreases, frequencies increase.

- **Conicity Effects:**

- Tapered bores show non-linear frequency shifts
- Mode spacing deviates from harmonic series
- Truncation reveals local vs. global geometry effects

- **End Correction:**

- Effective length includes end corrections
- Truncation point acts as new open end
- Correction depends on truncation diameter

Applications:

- Design optimization for specific frequency ranges
- Understanding bore geometry influence
- Comparative analysis across different flutes

Parameter Variation Studies

Concept: Systematically vary geometric parameters and observe acoustic response changes.

Available Parameters:

① **Hole Undercut Angle** (Priority 1)

- Modifies conical hole geometry
- Range: 0° (cylindrical) to 15° (strong undercut)
- Affects: Intonation, register response

② **Part Taper Angle** (Priority 2)

- Modifies bore conicity of individual parts
- Affects: Overall bore profile, mode spacing

③ **Wall Thickness** (Priority 3)

- Constant, variable, or proportional profiles
- Affects: External geometry, mass distribution

Sensitivity Analysis Workflow

Process:

- 1 **Select Base Flute:** Choose reference instrument
- 2 **Choose Parameter:** Select geometric parameter to vary
- 3 **Define Range:** Set min, max, and step values
- 4 **Generate Variants:** System creates geometric variants
- 5 **Calculate Acoustic Response:** Full impedance analysis for each variant
- 6 **Extract Metrics:** Compute all acoustic metrics
- 7 **Visualize Results:** Parameter vs. metric plots
- 8 **Generate Report:** Comprehensive PDF report

Output:

- Sensitivity plots (parameter vs. acoustic metrics)
- Complete acoustic analysis for all variants
- Statistical summaries
- Design recommendations

Interactive dialog for parameter selection and range specification

Sensitivity Analysis: Example Results

Typical Outputs:

- **Parameter Sweep Plots:**

- Undercut angle vs. inharmonicity
- Undercut angle vs. MOC
- Undercut angle vs. B_I and ESPE

- **Optimal Range Identification:**

- Parameter values minimizing inharmonicity
- Values achieving target MOC
- Trade-offs between different metrics

- **Design Insights:**

- Critical parameters requiring precision
- Tolerant parameters with wide acceptable ranges
- Interaction effects between parameters

Example plots showing undercut angle vs. inharmonicity, MOC, etc.

SQLite Database Structure:

- **Flutes Table:** General information, metadata
- **Flute Geometry:** Internal geometry per part (JSON)
- **External Geometry:** External profiles (measured/parametric)
- **Impedance Calculation Params:** Calculation parameters (hash-based)
- **Impedance Results:** Cached impedance computations
- **Bore Geometry:** OpenWind-compatible geometry
- **Side Holes:** Hole definitions for OpenWind

Benefits:

- Fast retrieval of previous calculations
- Parameter change detection
- Efficient storage of large impedance datasets
- Multi-flute comparison capabilities

Statistical Reporting

Comprehensive Database Reports:

① Geometric Statistics:

- Bore diameter distributions
- Hole size statistics
- Length measurements
- Conicity analysis

② Acoustic Statistics:

- Inharmonicity distributions
- MOC statistics across flutes
- B_f and ESPE ranges
- Q-factor distributions

③ Comparative Analysis:

- Flute-to-flute comparisons
- Historical period analysis
- Maker-specific characteristics

④ Visualizations:

- Histograms, box plots, scatter plots
- Correlation matrices
- Multi-flute overlays

Automated PDF Generation:

- **Technical Drawings:**

- Multi-view projections
- Dimensioned drawings
- Cross-sections
- Hole detail views

- **Geometric Accuracy:**

- Scale-accurate representations
- Proper dimensioning
- Manufacturing-ready format

- **Customization:**

- Select parts to include
- Choose views
- Add annotations

Automated technical drawings with dimensions and cross-sections

CNC Manufacturing Support:

- **Automatic G-Code Generation:**

- Toolpath generation for bore machining
- Hole drilling sequences
- Multi-axis support

- **Parameters:**

- Tool diameter and type
- Feed rates and speeds
- Depth of cut
- Safety clearances

- **Output:**

- Standard G-code format
- Compatible with common CNC controllers
- Validation and simulation support

CNC toolpath generation with configurable parameters

Features:

- **Visual Editing:**

- Click-and-drag hole positions
- Adjust diameters interactively
- Modify measurements in real-time

- **Validation:**

- Real-time geometry checks
- Acoustic preview
- Error detection

- **Export:**

- Save modified geometry to JSON
- Load directly into analysis
- Version control support

Visual editor for click-and-drag modification of flute geometry

Comparative Analysis

Simultaneous Multi-Flute Analysis:

- **Load Multiple Flutes:**

- From database or JSON files
- Filter by maker, period, type
- Custom selection

- **Overlaid Visualizations:**

- All acoustic metrics on same plots
- Color-coded by flute
- Easy comparison

- **Summary Dashboard:**

- Radar charts for multi-metric comparison
- Statistical tables
- Quick identification of outliers

Applications:

- Historical instrument comparison
- Maker style analysis
- Design evolution studies
- Quality control

Impedance Computation:

• Geometry Preparation:

- Bore segments: $[x_s, x_e, r_s, r_e, \text{type}]$
- Side holes with chimney heights
- Fingering charts

• Physical Parameters:

- Speed of sound: $c(T) = 331.3\sqrt{1 + T/273.15}$
- Radiation impedance (unflanged/flanged)
- Wall losses (Webster-Lokshin model)

• Computation:

- Transmission line method
- Frequency: 100-3000 Hz (configurable)
- Step: 2 Hz (default)

Processing Pipeline:

- ① **Input:** JSON Files / Database
- ② **Processing:**
 - FluteData: Load and validate geometry
 - OpenWind: Compute impedance
- ③ **Analysis:**
 - FluteAnalyzer: Extract metrics
 - Generate visualizations
 - Generate reports

All steps are cached in database for efficiency

Key Steps:

- ① Load/validate geometry
- ② Prepare OpenWind inputs
- ③ Compute impedance
- ④ Extract metrics
- ⑤ Generate visualizations
- ⑥ Export reports

Performance Optimizations

Caching Strategy:

- **Database Caching:**

- Store impedance results with parameter hash
- Detect parameter changes
- Skip recalculation if parameters unchanged

- **Lazy Loading:**

- 3D visualization loads only when accessed
- Large datasets loaded on demand
- Memory-efficient operation

- **Parallel Processing:**

- Multi-flute analysis can be parallelized
- Sensitivity analysis variants computed independently

Typical Performance:

- Single flute analysis: 10-30 seconds
- Multi-flute comparison: Scales linearly
- Database retrieval: $\leq$ 1 second (cached)

Academic Research:

- **Historical Instrument Analysis:**

- Compare instruments from different periods
- Study maker-specific characteristics
- Document acoustic evolution

- **Design Studies:**

- Understand geometry-acoustics relationships
- Optimize for specific acoustic goals
- Validate theoretical models

- **Player-Instrument Interaction:**

- Quantify compensation requirements
- Study register transition challenges
- Analyze playing difficulty metrics

Practical Applications

Instrument Making:

- **Design Optimization:**

- Sensitivity analysis guides design choices
- Identify critical vs. tolerant parameters
- Reduce trial-and-error iterations

- **Quality Control:**

- Compare new instruments to references
- Identify manufacturing deviations
- Ensure consistency

- **Reproduction:**

- Accurate geometric modeling
- Engineering drawings for manufacturing
- G-code for CNC machining

Education:

- Teaching acoustic principles
- Demonstrating geometric effects
- Interactive exploration tool

Potential Enhancements

Planned Features:

- **Advanced Analysis:**

- Modal analysis (beyond impedance minima)
- Time-domain simulations
- Nonlinear effects modeling

- **Machine Learning:**

- Predictive models for design optimization
- Pattern recognition in historical instruments
- Automated quality assessment

- **Extended Instrument Support:**

- Other woodwind instruments
- Brass instruments
- Custom geometries

- **Collaboration Features:**

- Cloud database sharing
- Collaborative analysis
- Version control integration

Summary

Key Strengths:

① Comprehensive Analysis:

- Multiple acoustic metrics
- Novel resonator truncation method
- Sensitivity analysis capabilities

② User-Friendly Interface:

- Intuitive PyQt5 GUI
- Interactive visualizations
- Detailed information buttons

③ Scientific Rigor:

- Based on established acoustic theory
- OpenWind integration
- Proper mathematical formulations

④ Practical Tools:

- Engineering drawings
- G-code generation
- Database management

Questions & Discussion

Software available for research collaboration

Contact: [Your contact information]